

The metaverse beckons

AS THE WORLD VENTURES TOWARDS A FUTURE WHERE EVERYTHING AROUND YOU HAS A CHIP IN IT AND IS CONNECTED TO THE INTERNET, there arises a massive cache of data that's constantly growing and reacting to its surroundings. In a way, this ecosphere of data is like our digital twin. We've got smart bands and smartwatches on our wrists tracking our vital stats on a minute-by-minute basis, and our smartphones are up to date with our daily schedules and even know where we will be at a certain point in time. Even if it's not penned into our schedules, it can certainly be projected on the basis of our past actions. Essentially, there's a "digital you" alive and performing all of your actions at the exact same time as you. It's just that we haven't put up a framework to visualise our digital twin.

Unless, of course, you're counting video games. In which case, we've been exploring new worlds using our digital twins for a couple of decades. However, the worlds in video games aren't analogous to the real world. Even games such as Second Life, which started off with the aim of replicating the real world ended up being exactly what our social media profiles turned out to be – fake. So, the search for a digital twin continued and it seems like it's soon going to end with the Metaverse.

The Metaverse is a shared collaborative space built to mimic real-world spaces and structures. Imagine if you had to go to work tomorrow, and all you had to do was wear a pair of AR or VR glasses and you'd be instantly transported to your virtual office. All of your colleagues, including that pesky person who's always trying to pawn off their work onto you, and the other person who makes it a point to microwave their tuna sandwich in the pantry. Except now, these individuals are no longer a bother and everyone's actions are visible to everyone else. The same could be true for classrooms which won't be limited to just the physical confines of the school building but could very well expand to show students clear visual representations of whatever they're learning. Every day could be a field trip to study the oxygen cycle or understand how humans came to be through the process of evolution.

While classrooms and more consumer oriented applications will always take time, businesses and enterprises would be the first to adopt such technologies. Certain businesses have already started simulating their factories in a digital environment to observe and optimise every aspect that they can right from the moment when the raw materials enter the premises to the point where the final product has been boxed and is ready to be shipped. The logistics of the raw materials before and after the factory are already simulated as part of the supply chain network. Which, by the way, wouldn't be difficult to integrate into their factory simulation.

Since we broached the topic of consumer-oriented applications and video games, we might as well talk about Fortnite. Epic's Chief Creative Officer, Donald Mustard, recently outlined plans for expanding the Fortnite universe to a much larger and equally ambitious story-telling experiment. Obviously, this would be a video game world which is still scripted to a large extent. Fortnite has already set massive benchmarks for digitising live events in the game. Music concerts held within Fortnite such as the one headed by Travis Scott was watched by 12.3 million viewers which puts every single concert held until now to shame.

Major tech companies such as NVIDIA have already been laying the building blocks for a couple of years already. NVIDIA is targeting enterprises with the Omniverse, which is their version of the Metaverse. They even simulated their new office building on their platform to optimise for cooling and lighting costs. The fact is that a lot of the thought process as well as the implementations have begun, albeit in a fragmented manner. Once standards for interconnects have been put in place, we'll see these fragmented metaverses come together to become The Metaverse, or if you prefer, the Multiverse. At least that's what I hope will come to fruition in the next 20 years. **d**



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